

15  
08  
26

→ For time-frequency plane  
must know the family of filter we  
are going to use and the other  
DTFT level

→ For better spectral description → (1st 2nd) time plane  
we better filter (good response in freq.)

→ For better time location →  
the suggests the filter is the  
want the time location will  
be. More is good for that

is linear  
in phase due to its linearity

→ For both we can → use a filter  
of regular size and decomposition  
but with intermediate value

→ Due to maladaptation the some  
vel. are shifted to the right (not in phase).  
So in some cases it can cause more  
time location

### - - - Application I: the time-frequency plane - - -

► It is as follows:

- Apply **individually**, for each leaf sub-band contained in the wavelet decomposition tree at the level  $j$ , a circular right-sided shift of  $\frac{N-2}{2}$  samples. Whenever this value exceeds the size of the sub-band, the shift is of  $(\frac{N-2}{2}) \% (\frac{M}{2^j})$  samples, where  $\%$  is the integer remainder of division, as in C/C++ language. Consequently, if  $\frac{N-2}{2}$  is a multiple of  $\frac{M}{2^j}$ , the correction is not required.
- That correction comes from the fact that the convolution between  $s[\cdot]$  and a wavelet filter produces a signal for which the length corresponds to  $M + N - 1$ , primarily. Then, once the downsampling by 2 is applied, this length is reduced to  $\frac{M+N-1}{2} = \frac{M}{2} + \frac{N-1}{2}$ . Lastly, the wraparound procedure vanishes the effect of the term  $\frac{N-1}{2}$ , remaining only a  $\frac{M}{2}$ -sample long resulting signal. Thus, even though  $N$  has no influence over the length of the transformed signal, it does cause shifts.

Due to this shift, maybe the pts disappeared when we plot the relation of the plot on

→ We can apply some correction to undo the shift effect. For large  $N$ , it's not the right way, but still can be done

→ Energy within  $\omega$  must be a scalar  
4. Though energy operator is given another signal instead of a scalar, we can transform it giving a energy signal. It reflects the physical effect the same did to create this signal.

it's band or - meaning  
system

- Specifically, due to the fact that  $N$  is usually even and half-shifts are not defined in discrete wavelet systems,  $\frac{N-1}{2}$  can be treated as being  $\frac{N-2}{2}$ , implying that all the filters but Haar's cause a shifting. As the decomposition advances from one level to the next, the shifting doubles, however, downsampling and wraparound cancel the doubling. Therefore, the procedure consists of a circular right-sided shift of  $\frac{N-2}{2}$  samples, regardless of  $j$ .
- Finally, a refined time-frequency description based on the above-mentioned strategies requires the use of Teager's energy operator, instead of the ordinary energy, to convert all the samples of each leaf sub-band into their instantaneous potentials, as follows.
- For the sub-band  $b = 0, 1, \dots, 2^j - 1$  of the deepest level  $j$ , do:
  - $\tilde{y}_{0,b} \leftarrow (y_{0,b})^2$ ;
  - $\tilde{y}_{i,b} \leftarrow (y_{i,b})^2 - (y_{i-1,b}) \cdot (y_{i+1,b})$ , for  $(1 \leq i \leq \frac{M}{2^j} - 2)$ , whenever the length of  $b$  is at least four;
  - $\tilde{y}_{\frac{M}{2^j}-1,b} \leftarrow (y_{\frac{M}{2^j}-1,b})^2$ , whenever the length of  $b$  is at least two.

Using the radar interface, only amplitude is taken into account. But with Teager's it measures both the amplitude and frequency contribution.

- ▶ Summarizing, the ideal time-frequency analysis of  $s[\cdot]$  based on the DTWT requires the following steps:

- ▶ **STEP 1:** Select the most appropriate wavelet family and the support-size of filters, i.e.,  $N$ . Then, calculate the  $M$ -sample long transformed signal  $y[\cdot]$  from  $s[\cdot]$  based on the DTWT, adequately choosing the transformation level  $j$ ;
- ▶ **STEP 2:** Apply the correction described above related to the rotations, modifying  $y[\cdot]$  accordingly;
- ▶ **STEP 3:** Convert each tree leaf into its Teager energy signal, creating  $\tilde{y}[\cdot]$  from rotated  $y[\cdot]$  and, lastly, interpret  $\tilde{y}[\cdot]$ .

- ▶ For a better understanding, we will take a look at four examples: the first, second and third can be followed by hand whereas the fourth results from a computational implementation involving a long input.

- ▶ **Numerical Example 1:** Letting  $s[\cdot] = \{3, -2, 6, -1, 4, -2, 2, -1\}$ , which is part of a long signal sampled at 44100 samples per second, identify its predominant spectral component and the corresponding time-support, aiming at maximum time resolution.

↳ In the graph we want precise of the location, but here is spectral resolution

we need to know the range not

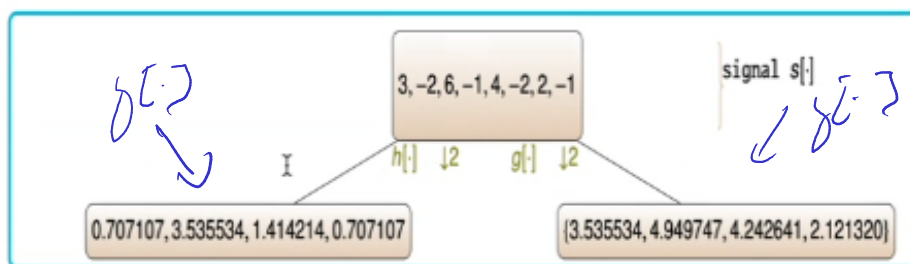


- **Solution:** We wish maximum time resolution. This means we want the most precise information about where the dominant frequency is localized within  $s[\cdot]$ . In consequence, as a result from Heisenberg's uncertainty principle, the spectral resolution will not be so good, i.e., we will only be able to state that the dominant frequency is something within a big range.
- From STEP 1 in the previous slide, we need a decision about the wavelet family to be used and the corresponding support-size of the filters. If a maximum time resolution is the priority, then the filter support-size must be the smallest. This is because the bigger the support-size of the filters is, the wider the portion of  $s[\cdot]$  captured by each sample of the transformed signal will be. Therefore, we adopt **Haar** filters to solve this problem! In addition, the lower the transformation level  $j$  is, the better the time resolution will be. Thus, we will perform the first level transformation, i.e.,  $j = 1$ .

*Haar is the best & much easier*

*the chosen Haar filters*

The corresponding decomposition tree is:



*left branch*

- From STEP 2, we learned that a right-sided shift of  $\frac{N-2}{2}$  samples is required in each tree leaf. But since for Haar filters  $N = 2$ , then,  $\frac{N-2}{2} = \frac{2-2}{2} = 0$ , i.e., no rotation is required.
- From STEP 3, we learned that Teager Energy Operator provides a better description of each leaf energy. That operator does not produce a scalar value, as the ordinary energy. Instead, it produces another signal, i.e., an energy signal  $\tilde{y}[\cdot]$ . Thus, the calculations are as follows.

- - - Application I: the time-frequency plane - - -

- For the trend, which corresponds to the left child leaf, i.e.,  $\{0.707107, 3.535534, 1.414214, 0.707107\}$ , we have:

►  $\tilde{y}_{0\text{trend}} = (0.707107)^2 = 0.50$

$$\blacktriangleright \tilde{y}_{1\text{trend}} = (3.535534)^2 - 0.707107 \cdot 1.414214 = 11.50$$

$$\blacktriangleright \hat{y}_{2\text{trend}} = (1.414214)^2 - 3.535534 \cdot 0.707107 = -0.50$$

►  $\hat{y}_{3\text{trend}} = (0.707107)^2 = 0.50$

Thus,  $\tilde{y}_{\text{trend}}[\cdot] = \{0.50, 11.50, -0.50, 0.50\}$ .

- For the fluctuation, which corresponds to the right child leaf, i.e.,  $\{3.535534, 4.949747, 4.242641, 2.121320\}$ , we have:

►  $\tilde{y}_{\text{fluctuation}} = (3.535534)^2 = 12.50$

$$\text{► } \tilde{y}_{1\text{fluctuation}} = (4.949747)^2 - 3.535534 \cdot 4.242641 = 9.50$$

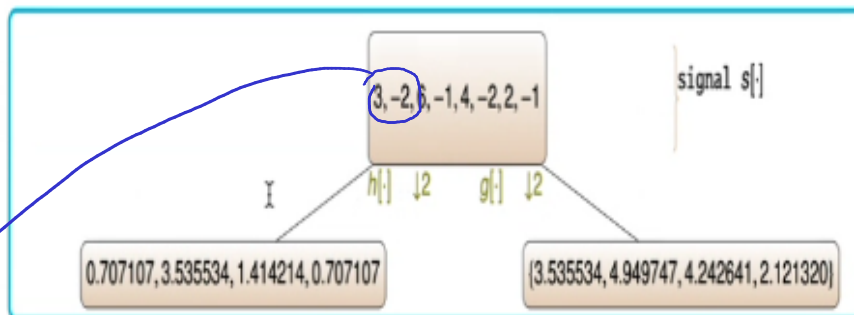
$$\text{► } \bar{y}_{2\text{fluctuation}} = (4.242641)^2 - 4.949747 \cdot 2.121320 = 7.50$$

►  $\hat{y}_{3\text{fluctuation}} = (2.121320)^2 = 4.50$

Thus,  $\tilde{y}_{\text{fluctuation}}[\cdot] = \{12.50, 9.50, 7.50, 4.50\}$ .

↑  $(len)^2$  - previous rest  
if there's still 19 previous & rest,  
it's 0 and only  $(len)^2$  remains  
we then use  $\approx [ ]$  for our analysis

The corresponding decomposition tree is:



- ▶ From STEP 2, we learned that a right-sided shift of  $\frac{N-2}{2}$  samples is required in each tree leaf. But since for Haar filters  $N = 2$ , then,  $\frac{N-2}{2} = \frac{2-2}{2} = 0$ , i.e., no rotation is required.
- ▶ From STEP 3, we learned that Teager Energy Operator provides a better description of each leaf energy. That operator does not produce a scalar value, as the ordinary energy. Instead, it produces another signal, i.e., an energy signal  $\tilde{y}[\cdot]$ . Thus, the calculations are as follows.

bring that 12.5 is the highest value, this is where the frequency starts to be detected and it's frequency range from 11028 - 22056 Hz (phonetic part)

- - - Application I: the time-frequency plane - - -

- **Numerical Example 2:** Letting  $s[\cdot] = \{4, 6, 10, 12, 8, 6, 5, 5\}$ , which is part of a long signal sampled at 16000 samples per second, identify its predominant spectral component and the corresponding time-support, aiming at a maximum frequency resolution.
- **Solution:** We need maximum frequency resolution. Thus, we want the most precise information about the dominant frequency. Consequently, just a vague information on where it is localized within  $s[\cdot]$  will be available.
- From STEP 1 defined in slide 122, we need a decision about the wavelet family to be used and the corresponding support-size of the filters. Since highest frequency resolution is the priority, the filter support-size must be as big as possible. This is because the bigger the support-size of the filters is, the closer to the ideal response its frequency response will be. Theoretically, wavelet filters do not have a maximum support-size, however, in practice, support-sizes around 30 or 40 are considered sufficient in such a case.

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is Maximal filter as we need less than 70 samples, because it's impractical to solve those equations



## Wavelet Transform and its Applications

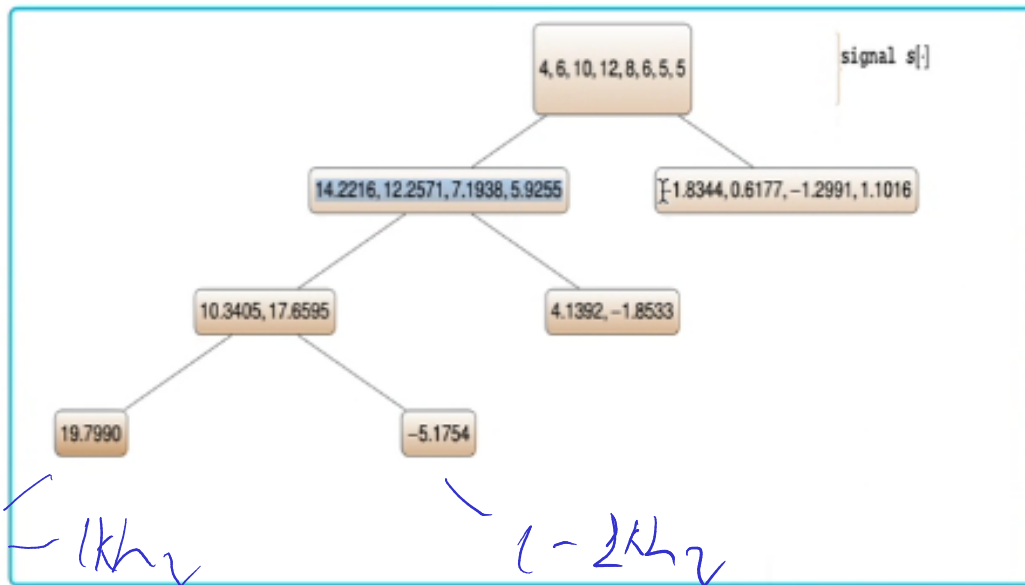
- - - Application I: the time-frequency plane - - -

- In addition, phase responses as linear as possible are desired to avoid distortion in the filtered signals. Consequently, Symmlets or Coiflets are usually preferable. Therefore, we adopt Coif-30 here and use the deepest decomposition level, i.e.,  $\log_2 8 = 3$  in this case. The decomposition tree is as follows.

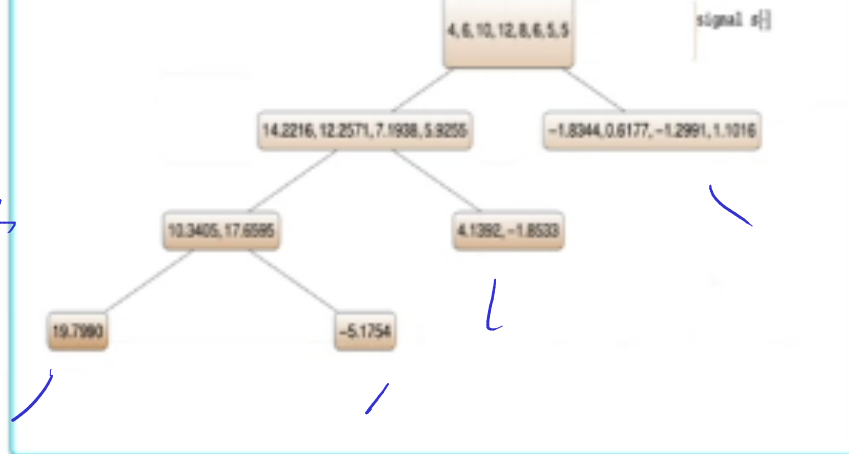
we're going to design a much as we can



In addition, phase responses as linear as possible are desired to avoid distortion in the filtered signals. Consequently, Symmlets or Coiflets are usually preferable. Therefore, we adopt Coif-30 here and use the deepest decomposition level, i.e.,  $\log_2 8 = 3$  in this case. The decomposition tree is as follows.



only leaves



127/200

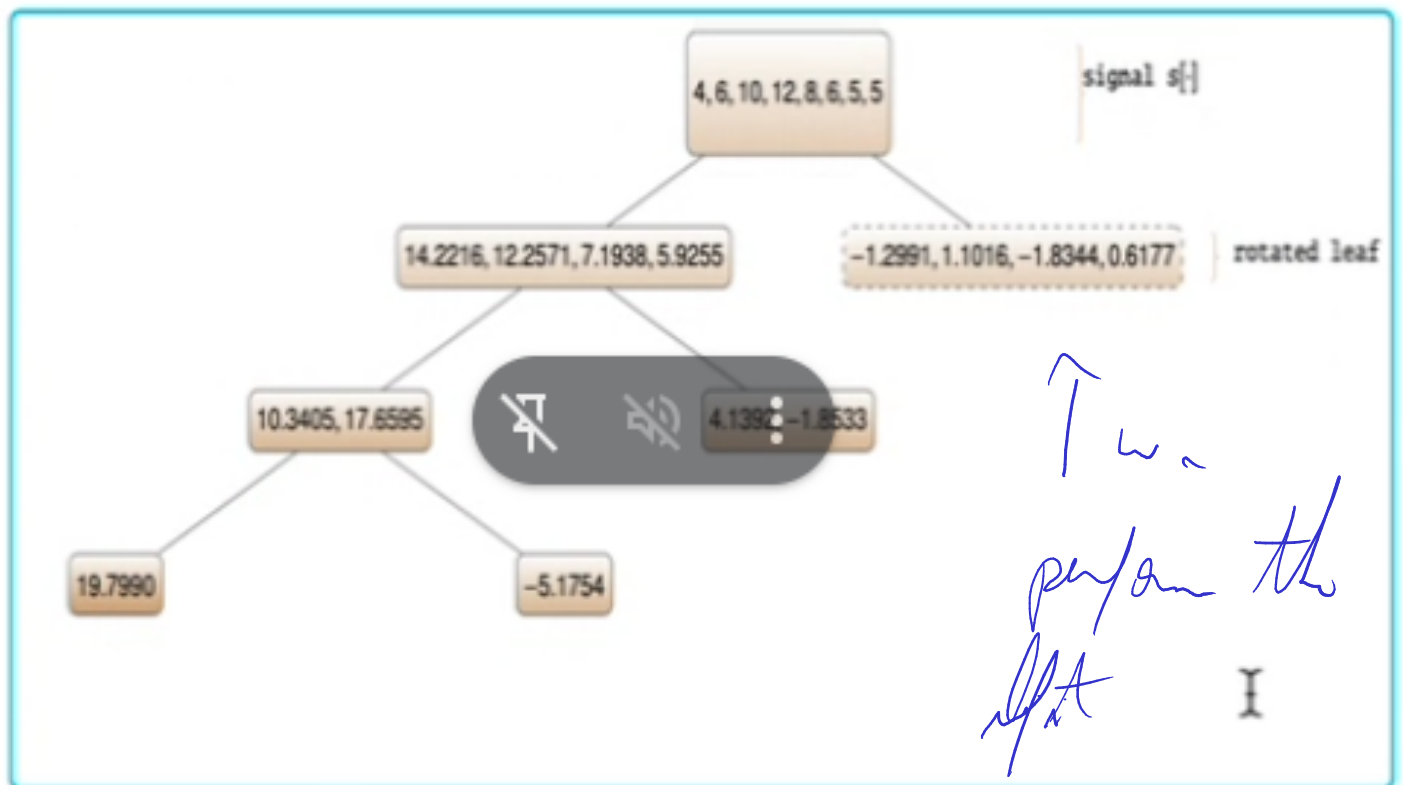


## Wavelet Transform and its Applications

- - - Application: the time-frequency plane - - -

- From STEP 2, we learned that a minor adjustment corresponding to a right-sided shift might be required in each tree leaf for a more precise time-frequency mapping. Particularly,
  - for both the leaves in the third level, the rotation is of  $\left(\frac{N-2}{2}\right)\% \left(\frac{M}{2^l}\right) = \left(\frac{30-2}{2}\right)\% \left(\frac{8}{2^3}\right) = 14\%1 = 0$  samples, i.e., no rotation is required.
  - for the leaf in the second level, the rotation is of  $\left(\frac{N-2}{2}\right)\% \left(\frac{M}{2^l}\right) = \left(\frac{30-2}{2}\right)\% \left(\frac{8}{2^2}\right) = 14\%2 = 0$  samples, i.e., no rotation is required.
  - for the leaf in the first level, the rotation is of  $\left(\frac{N-2}{2}\right)\% \left(\frac{M}{2^l}\right) = \left(\frac{30-2}{2}\right)\% \left(\frac{8}{2^1}\right) = 14\%4 = 2$  samples.
- Thus, considering the rotations, we have the following elements in the tree:

the shift go left by leaf indexing



### - - - Application I: the time-frequency plane - - -

#### ► In the third level:

- $\tilde{y}_{0\text{trend.3rd.level}} = (19.7990)^2 = 392$
- $\tilde{y}_{0\text{fluctuation.3rd.level}} = (-5.1754)^2 = 26.78$

#### ► In the second level:

- $\tilde{y}_{0\text{fluctuation.2nd.level}} = (4.1392)^2 = 17.13$
- $\tilde{y}_{1\text{fluctuation.2nd.level}} = (-1.8533)^2 = 3.43$

#### ► In the first level:

- $\tilde{y}_{0\text{fluctuation.1st.level}} = (-1.2991)^2 = 1.68$
- $\tilde{y}_{1\text{fluctuation.1st.level}} = (1.1016)^2 - (-1.2991) \cdot (-1.8344) = -1.17$
- $\tilde{y}_{2\text{fluctuation.1st.level}} = (-1.8344)^2 - 1.1016 \cdot 0.6177 = 2.68$
- $\tilde{y}_{3\text{fluctuation.1st.level}} = (0.6177)^2 = 0.38$

Is energy we consider the separate lines

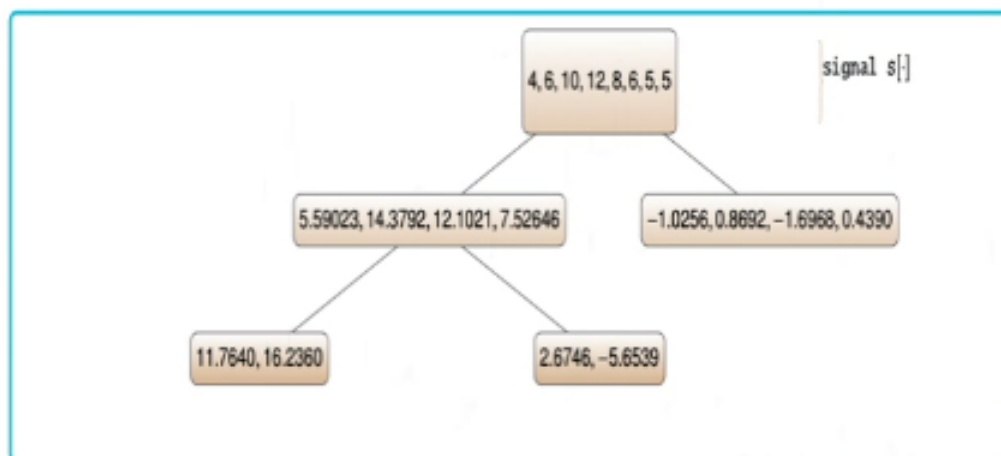




### - - - Application I: the time-frequency plane - - -

- ▶ **Numerical Example 3:** Let's repeat the previous example, considering the same input signal  $s[\cdot] = \{4, 6, 10, 12, 8, 6, 5, 5\}$  of size  $M = 8$  and sampled at 16000 samples per second, but aiming at the identification of its predominant spectral component with a balanced temporal and spectral resolutions.
- ▶ **Solution:** To get a balanced temporal and spectral resolutions, we use the medium decomposition level. Considering that the minimum is always 1 and the maximum is  $\log_2(M) = \log_2(8) = 3$ , the mean is  $j = \frac{1+3}{2} = 2$ .
- ▶ From STEP 1 defined in slide 122, we need once again a decision about the wavelet family and the corresponding support-size of the filters. Regarding the latter, we know that the minimum support is 2 and there is no theoretical limit. Since we considered that supports around 30 could be used in practice as a good limit for best filtering, we can use a medium support, i.e., a value around  $\frac{2+30}{2} = 16$ . Keep in mind that this is, however, a flexible possibility.

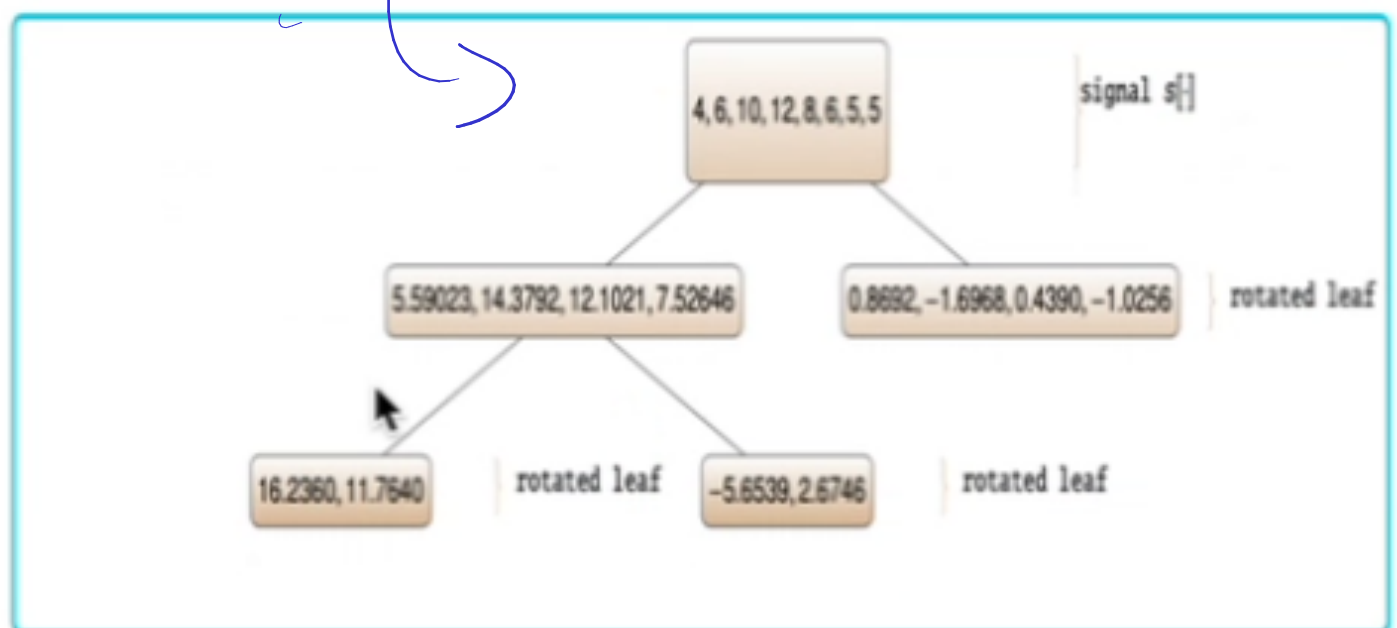
- ▶ Accordingly, phase responses as linear as possible are desired to avoid distortion in the filtered signals. Consequently, Symmlets or Coiflets are usually preferable. Therefore, let's adopt Symm-16 this time together with a medium decomposition level, i.e., 2. The decomposition tree is as follows.



- From STEP 2, we learned that a minor adjustment corresponding to right-sided shift might be required in each tree leaf for a more precise time-frequency mapping.

Particularly,

- for the leaves in the second level, the rotation is of  $\left(\frac{N-2}{2}\right)\% \left(\frac{M}{2^l}\right) = \left(\frac{16-2}{2}\right)\% \left(\frac{8}{2^2}\right) = 7\%2 = 1$  sample.
- for the leaf in the first level, the rotation is of  $\left(\frac{N-2}{2}\right)\% \left(\frac{M}{2^l}\right) = \left(\frac{16-2}{2}\right)\% \left(\frac{8}{2^1}\right) = 7\%4 = 3$  samples.
- Thus, considering the rotations, we have the following elements in the tree:



## - - - Application I: the time-frequency plane - - -

### ► In the second level:

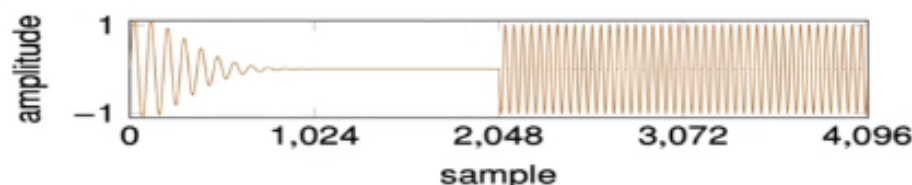
- $\tilde{y}_{0\text{trend.2nd.level}} = (16.2360)^2 = 263.61$
- $\tilde{y}_{1\text{trend.2nd.level}} = (11.7640)^2 = 138.39$
- $\tilde{y}_{0\text{fluctuation.2nd.level}} = (-5.6539)^2 = 31.97$
- $\tilde{y}_{1\text{fluctuation.2nd.level}} = (2.6746)^2 = 7.15$

### ► In the first level:

- $\tilde{y}_{0\text{fluctuation.1st.level}} = (0.8692)^2 = 0.76$
- $\tilde{y}_{1\text{fluctuation.1st.level}} = (-1.6968)^2 - 0.8692 \cdot 0.4390 = 1.32$
- $\tilde{y}_{2\text{fluctuation.1st.level}} = (0.4390)^2 - (-1.6968) \cdot (-1.0256) = -1.55$
- $\tilde{y}_{3\text{fluctuation.1st.level}} = (-1.0256)^2 = 1.05$

*dominant is the frequency is within 0~2kHz, in the first part of SFT*

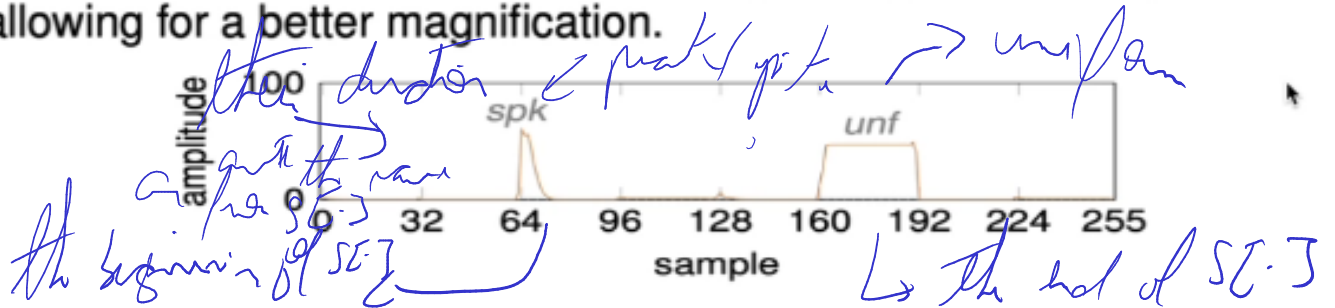
- **Computational Example:** Let  $s_i = 1.2\sin(\frac{2\pi \cdot 44i}{4095})e^{-\frac{i}{409.6}}$  for  $\{0 \leq i < 1024\}$ ,  $s_i = 0$  for  $\{1024 \leq i < 2048\}$  and  $s_i = \sin(\frac{2\pi \cdot 85i}{4095})$  for  $\{2048 \leq i < 4096\}$  to be the one-second long signal under analysis, sampled at  $2T = 4096$  samples per second, as shown in the Figure below. The problem is to find out the time-support of the main frequencies in  $s[\cdot]$  with a **balance** between temporal and spectral resolutions.



- **Solution:** A balanced time-frequency analysis of  $s[\cdot]$ , which contains frequencies up to  $T = 2048$  Hz, requires  $j = \left\lfloor \frac{1+\log_2(M)}{2} \right\rfloor = \left\lfloor \frac{1+\log_2(4096)}{2} \right\rfloor = 6$ , where  $\lfloor \cdot \rfloor$  is the floor operator. Symm-16 were chosen to perform the analysis as explained in the previous example.

- - - Application I: the time-frequency plane - - -

- The corresponding sixth-level DTWT, **processed and modified as in STEPS 2 and 3** previously defined in order to originate  $\tilde{y}[\cdot]$ , is shown in the next Figure. **Only the first 256 samples were plotted**, allowing for a better magnification.



- Observing the Figure, we can see two main elements: *spk* and *unf*. The former is a spike decaying from  $s_{64}$  to  $s_{79}$  and the latter consists of a uniform step from  $s_{160}$  to  $s_{191}$ . Clearly, *spk* reflects the exponential sinusoid in  $s[\cdot]$  and *unf* represents the uniform sinusoid in that input signal.